

Interim Revision – Visual Perception

Prof Anna Grubert

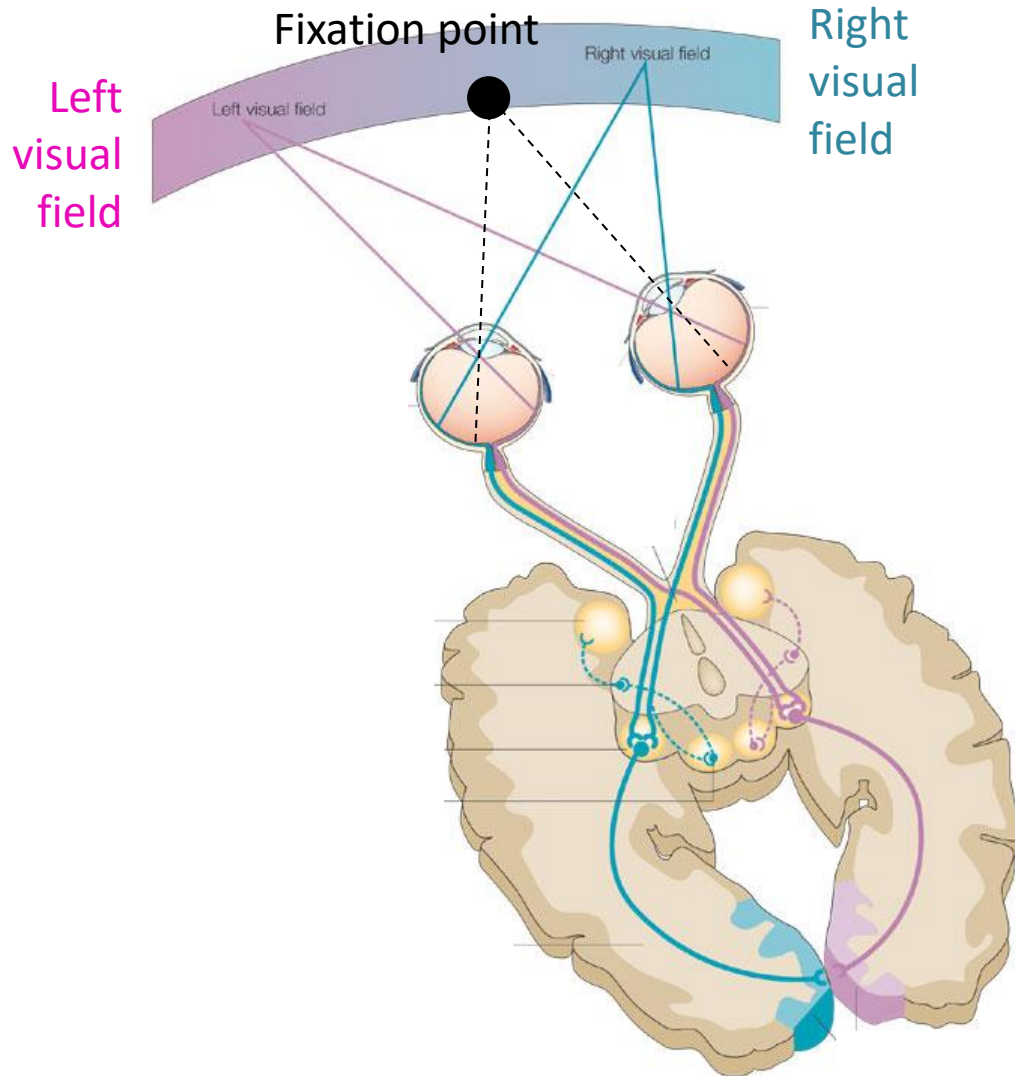
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Interim revision

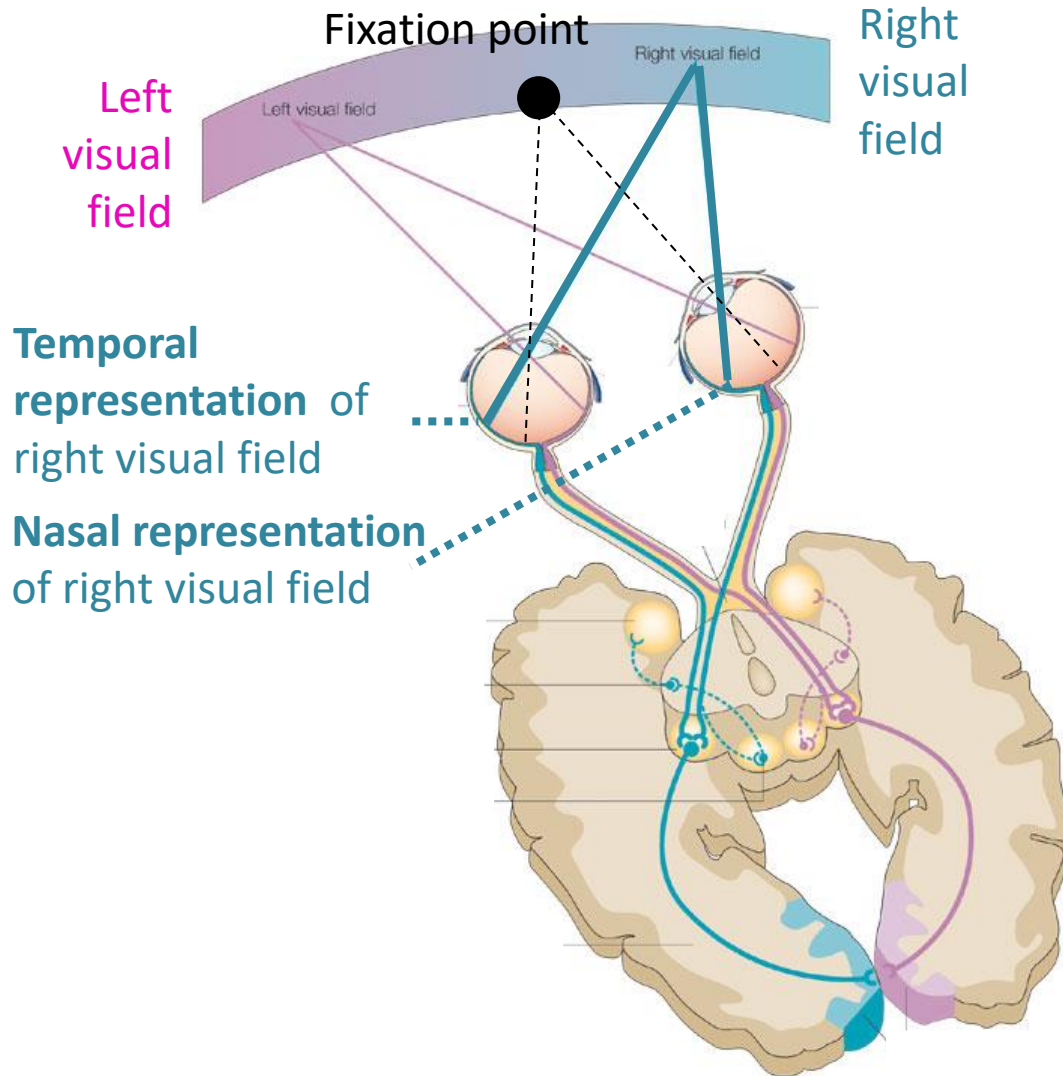
- Your questions were concerned with:
 1. The visual pathway (lecture 2)
 2. Specific neurons & their receptive fields (lecture 2, 5 & 7)
 3. The corollary discharge theory (lecture 7)
 4. Perception of mixed colours (lecture 5)
 5. Theories of perception (lectures 3 & 4)

1. The visual pathway

Retino-geniculo-striate pathway: right visual field

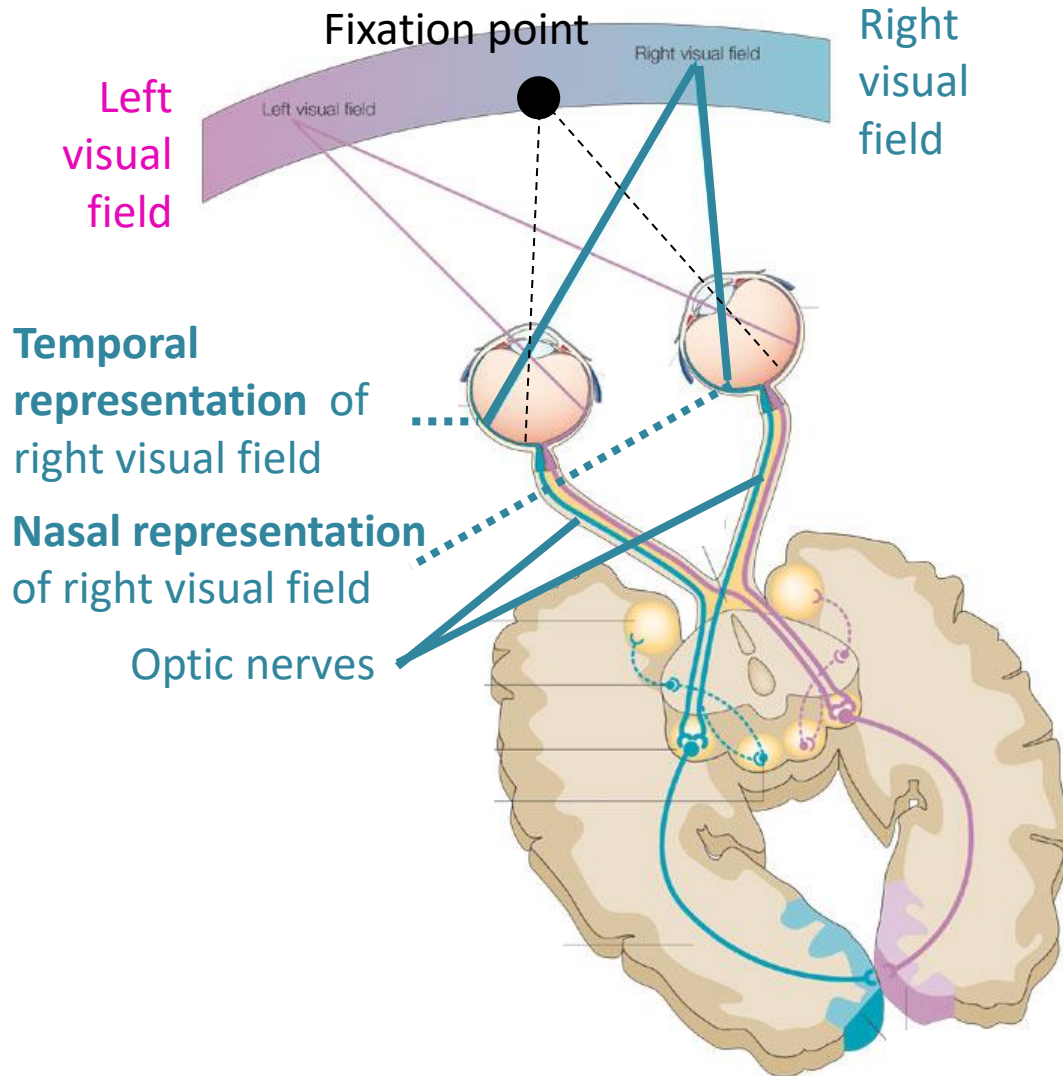


Retino-geniculo-striate pathway: right visual field



1. **Right visual field** is represented in the **nasal part of right eye** and in the **temporal part of left eye**.

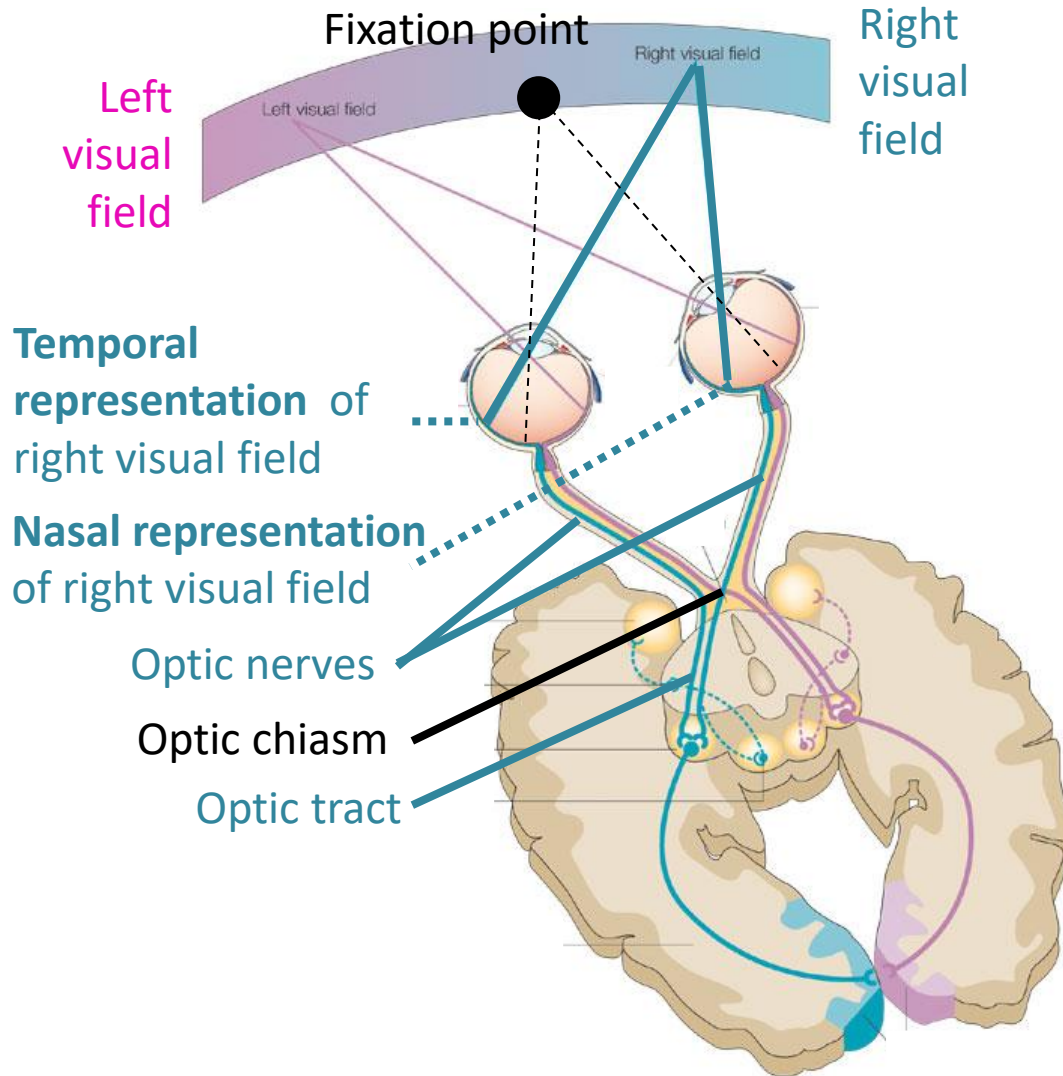
Retino-geniculo-striate pathway: right visual field



1. **Right visual field** is represented in the **nasal part of right eye** and in the **temporal part of left eye**.

2. Ganglion axons leave both eyes as **optic nerves**.

Retino-geniculo-striate pathway: right visual field

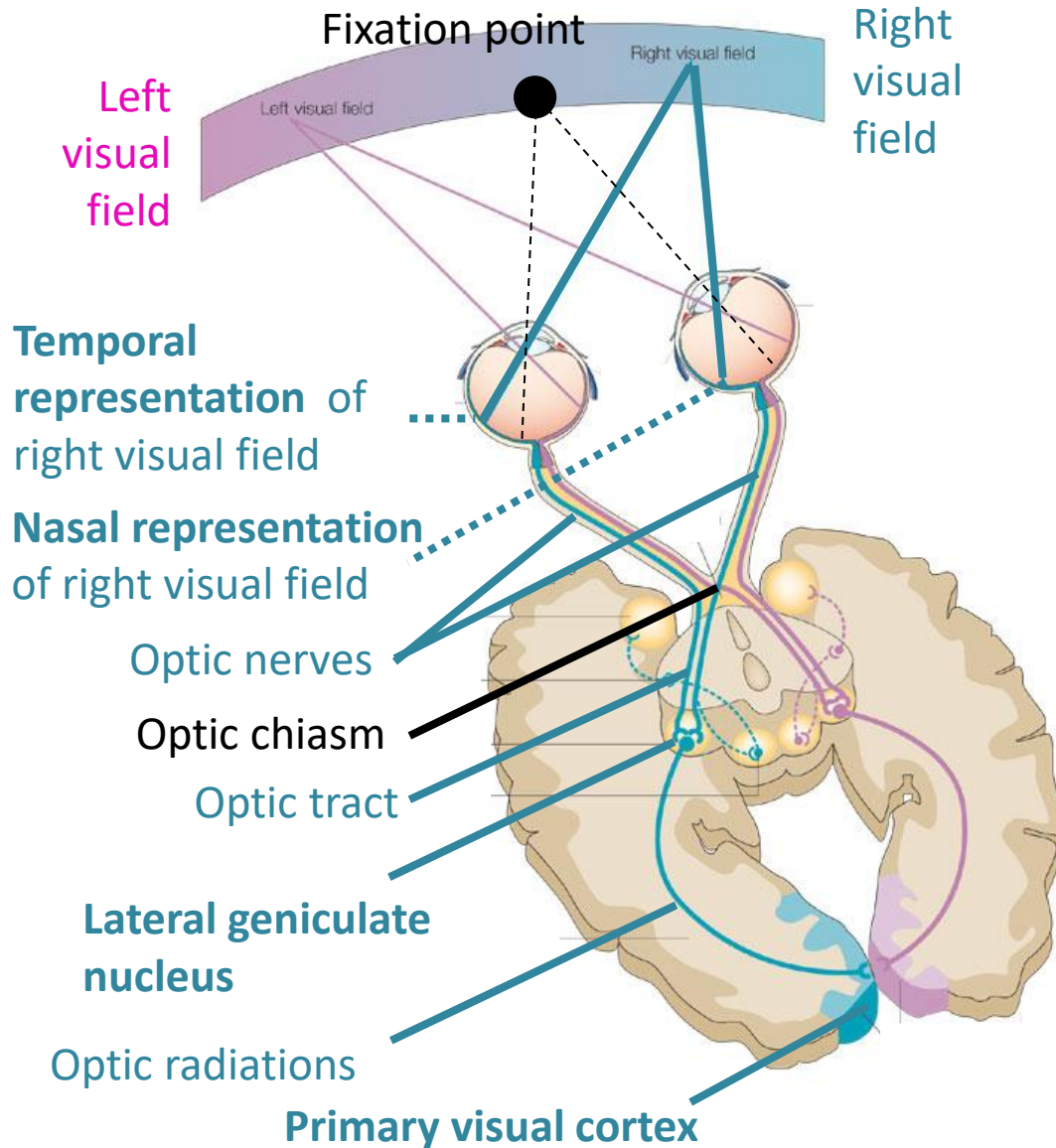


1. **Right visual field** is represented in the **nasal part of right eye** and in the **temporal part of left eye**.

2. Ganglion axons leave both eyes as **optic nerves**.

3. Nasal axons cross sides in the **optic chiasm** (nasal axons from the right eye go to the left side of the brain). All axons carrying right visual field information (nasal axons from right eye, temporal axons from left eye) combine to the left **optic tract**.

Retino-geniculo-striate pathway: right visual field



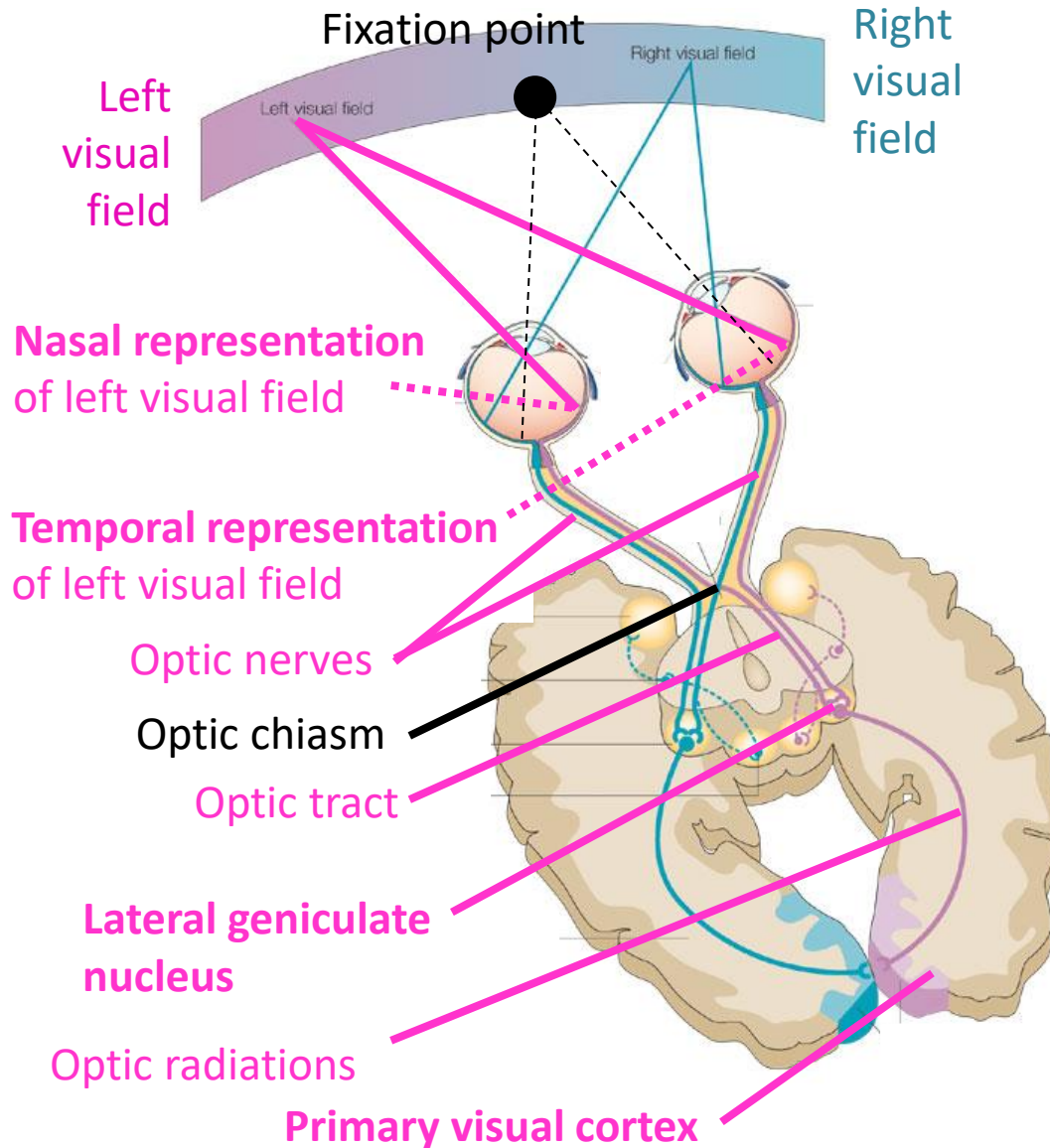
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2. Ganglion axons leave both eyes as **optic nerves**.

3. Nasal axons cross sides in the **optic chiasm** (nasal axons from the right eye go to the left side of the brain). All axons carrying right visual field information (nasal axons from right eye, temporal axons from left eye) combine to the left **optic tract**.

4. Left optic tract reaches the left **LGN**. Axons fan out in the left **optic radiations**, which go to left **V1**, which initially processes only right visual field information.

Retino-geniculo-striate pathway: **left visual field**



1. **Left visual field** is represented in the **nasal part of left eye** and in the **temporal part of right eye**.

2. Ganglion axons leave both eyes as **optic nerves**.

3. Nasal axons cross side in the **optic chiasm** (nasal axons from the left eye go to the right side of the brain). All axons carrying left visual field information (nasal axons from left eye, temporal axons from right eye) combine to the right **optic tract**.

4. Right optic tract reaches right **LGN**. Axons fan out in the right **optic radiations**, which go to right **V1**, which initially processes only left visual field information.

Retino-geniculo-striate pathway

Retina

Ganglion cells
with on₍₊₎/off₍₋₎
centre/surround
receptive fields.

M cells

M = Magnocellular

- Input from rods
- Colour-blind

P cells

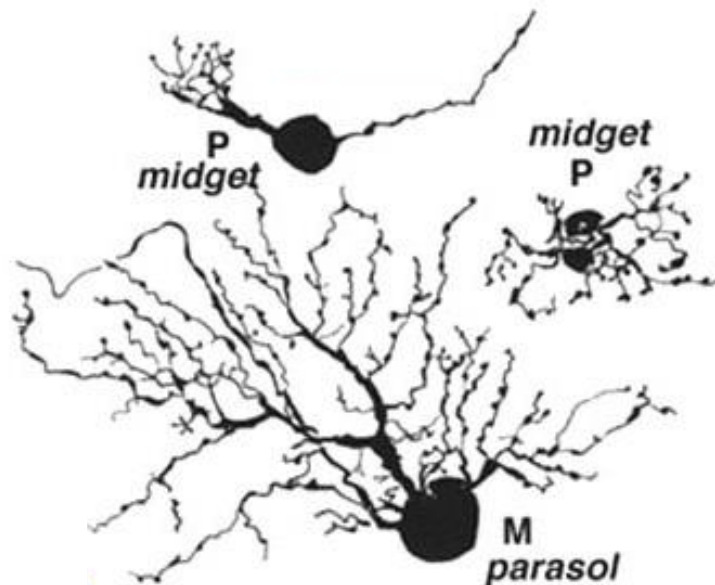
P = Parvocellular

- Input from M&L cones
- Green to red spectrum

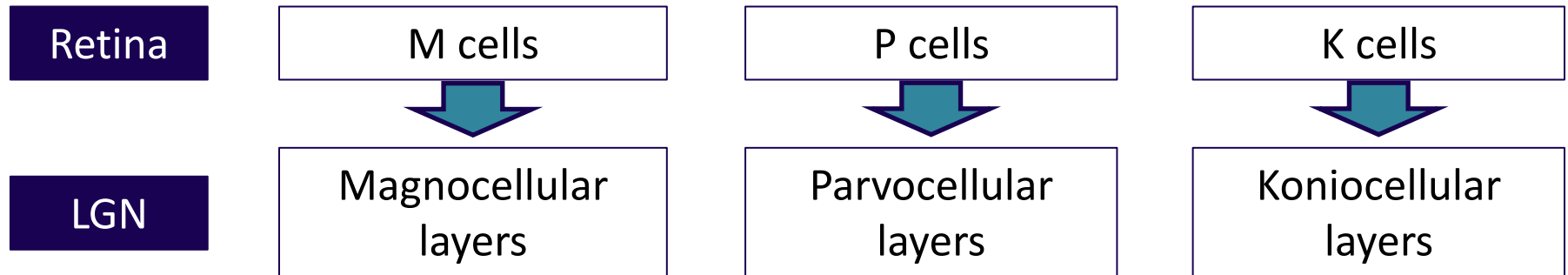
K cells

K = Koniocellular

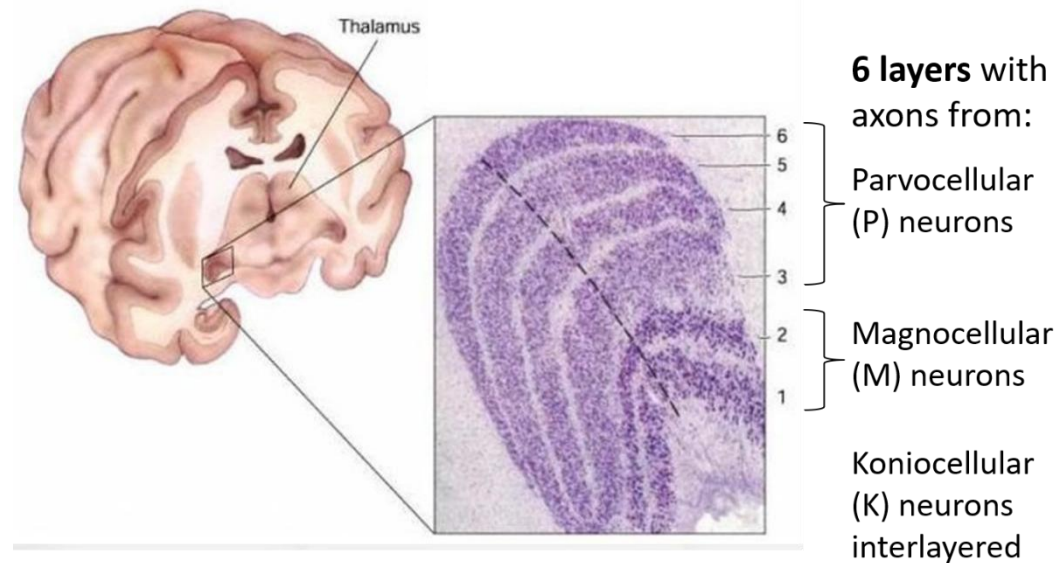
- Input from S cones
- Blue spectrum



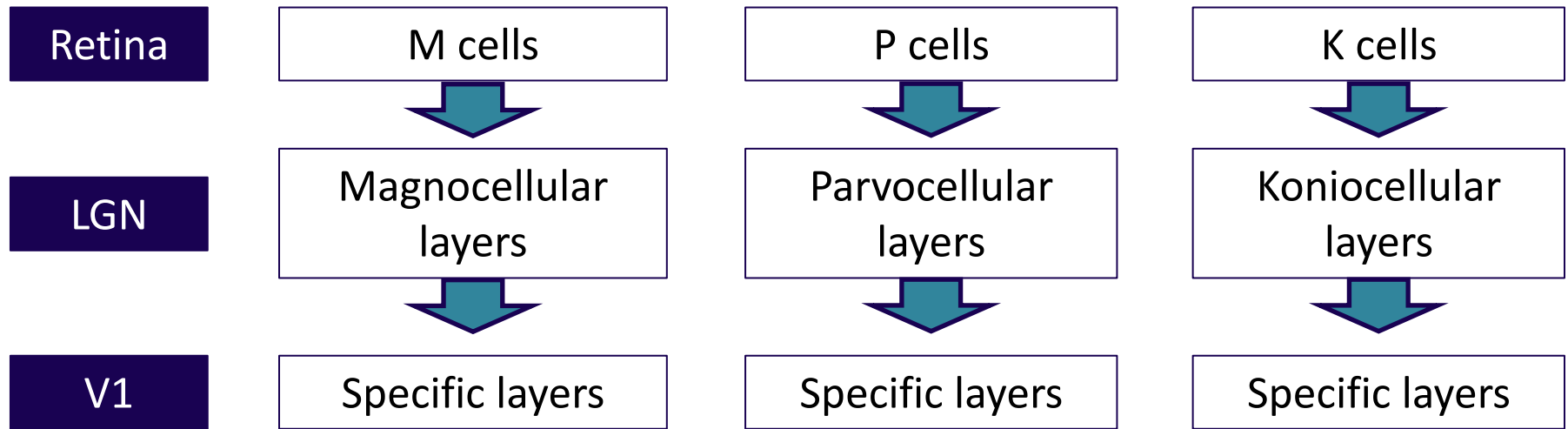
Retino-geniculo-striate pathway



LGN cells with
 $on_{(+)}/off_{(-)}$
centre/surround
receptive fields.

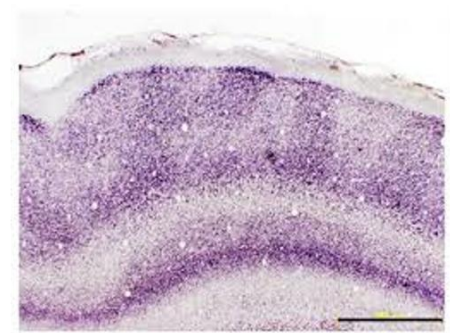
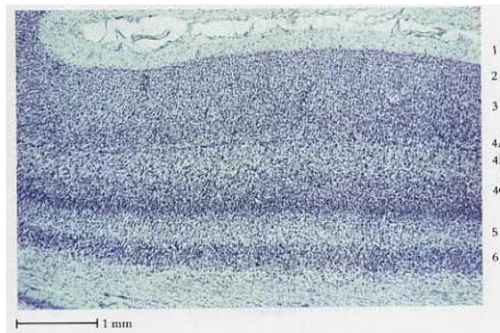


Retino-geniculo-striate pathway

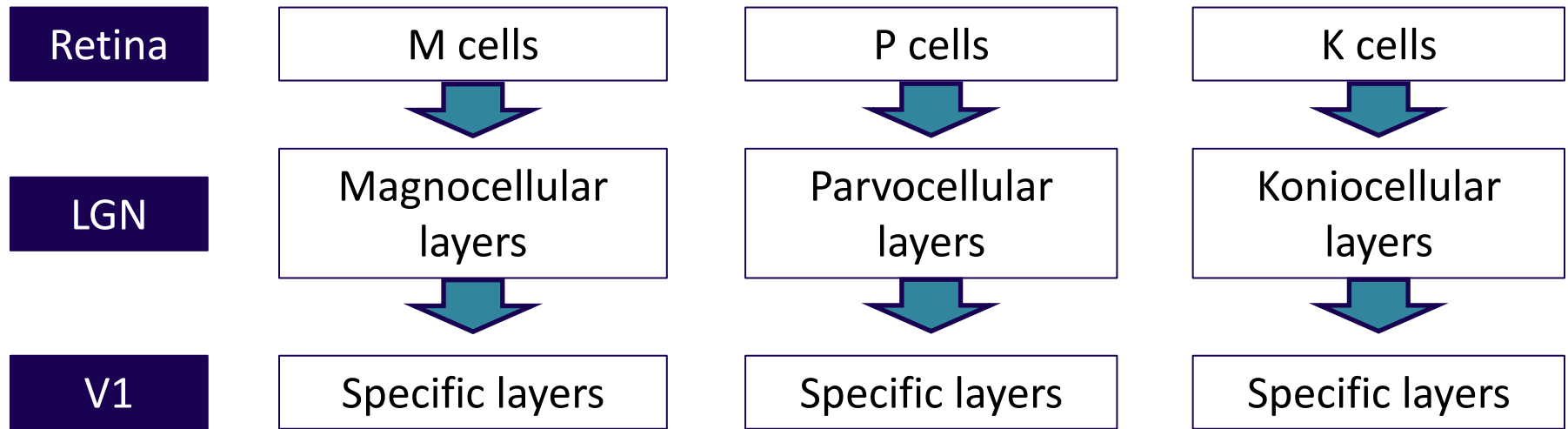


= Primary Visual Cortex = Striate Visual Cortex

Magno and parvo layers (from LGN and retina) are preserved in V1 input layers, but then merge subsequently.

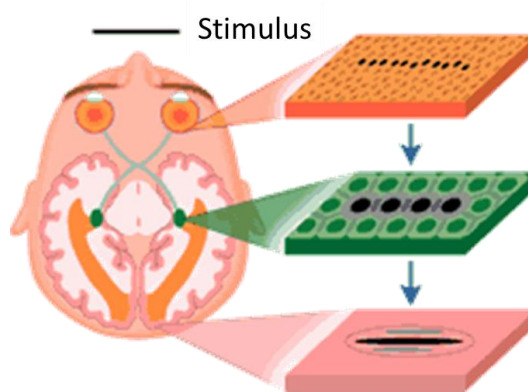


Retino-geniculo-striate pathway



= Primary Visual Cortex = Striate Visual Cortex

Simple cortical cells with **side-by-side** receptive fields for orientation & motion direction selective cells, or with **centre/surround** receptive fields for colour & brightness selective cells.

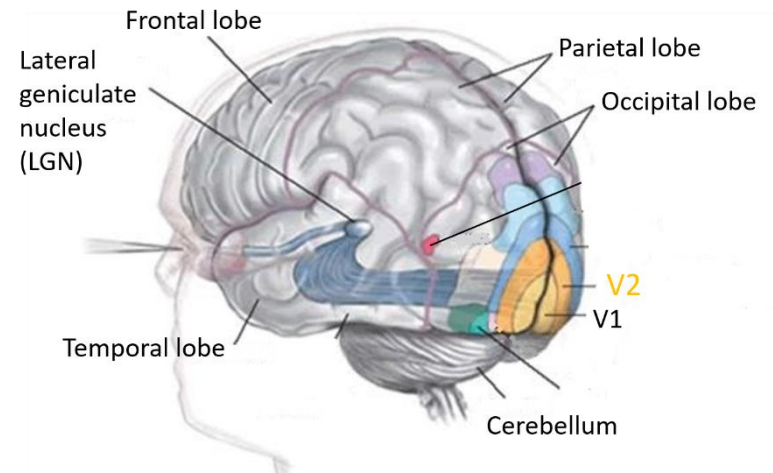
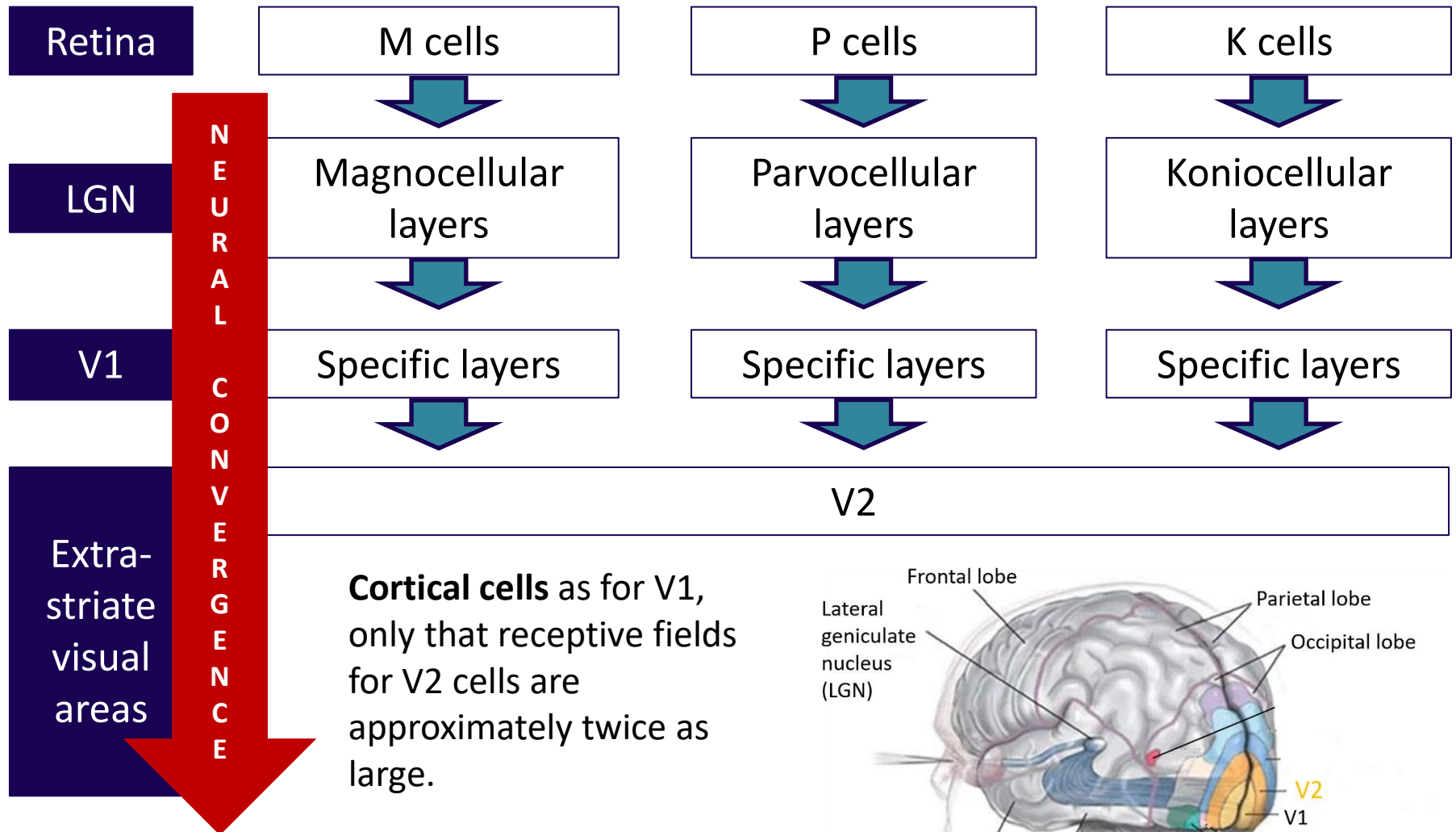


Ganglion cells in retina:
Centre-surround receptive fields.

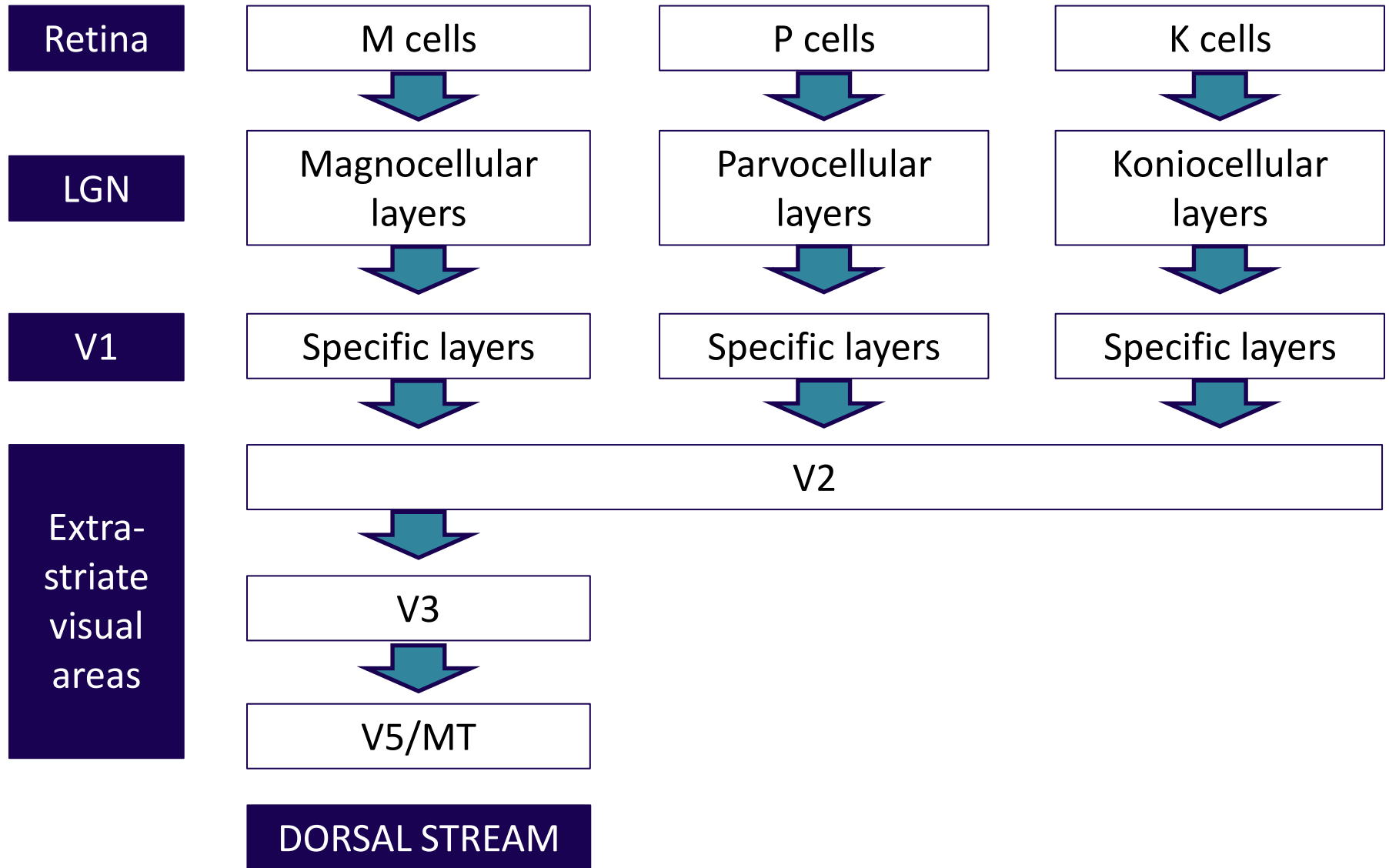
Lateral geniculate nucleus (LGN):
Centre-surround receptive fields.

Primary visual cortex (V1):
Simple cortical cells, side-by-side receptive fields.

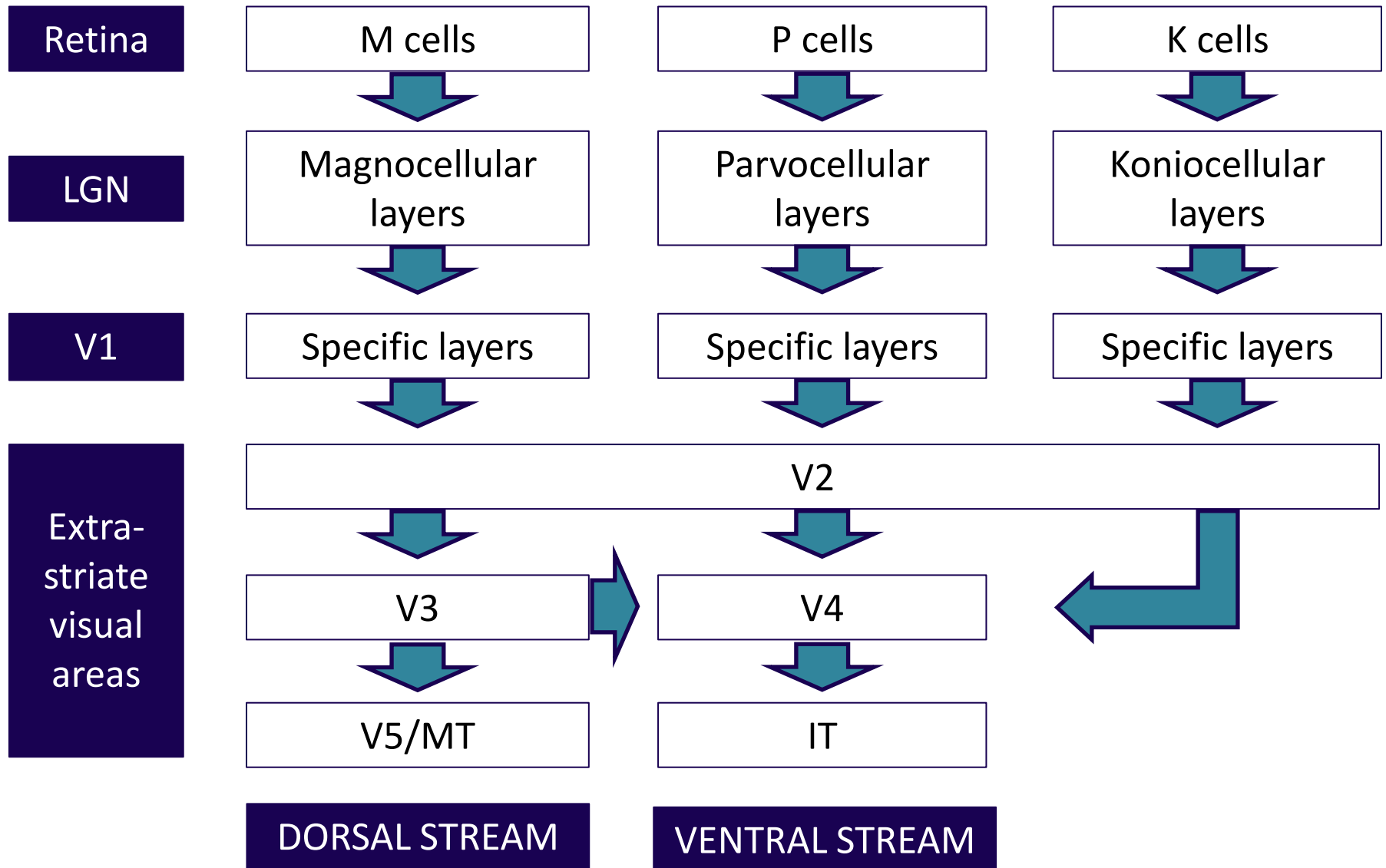
Cortical visual pathways



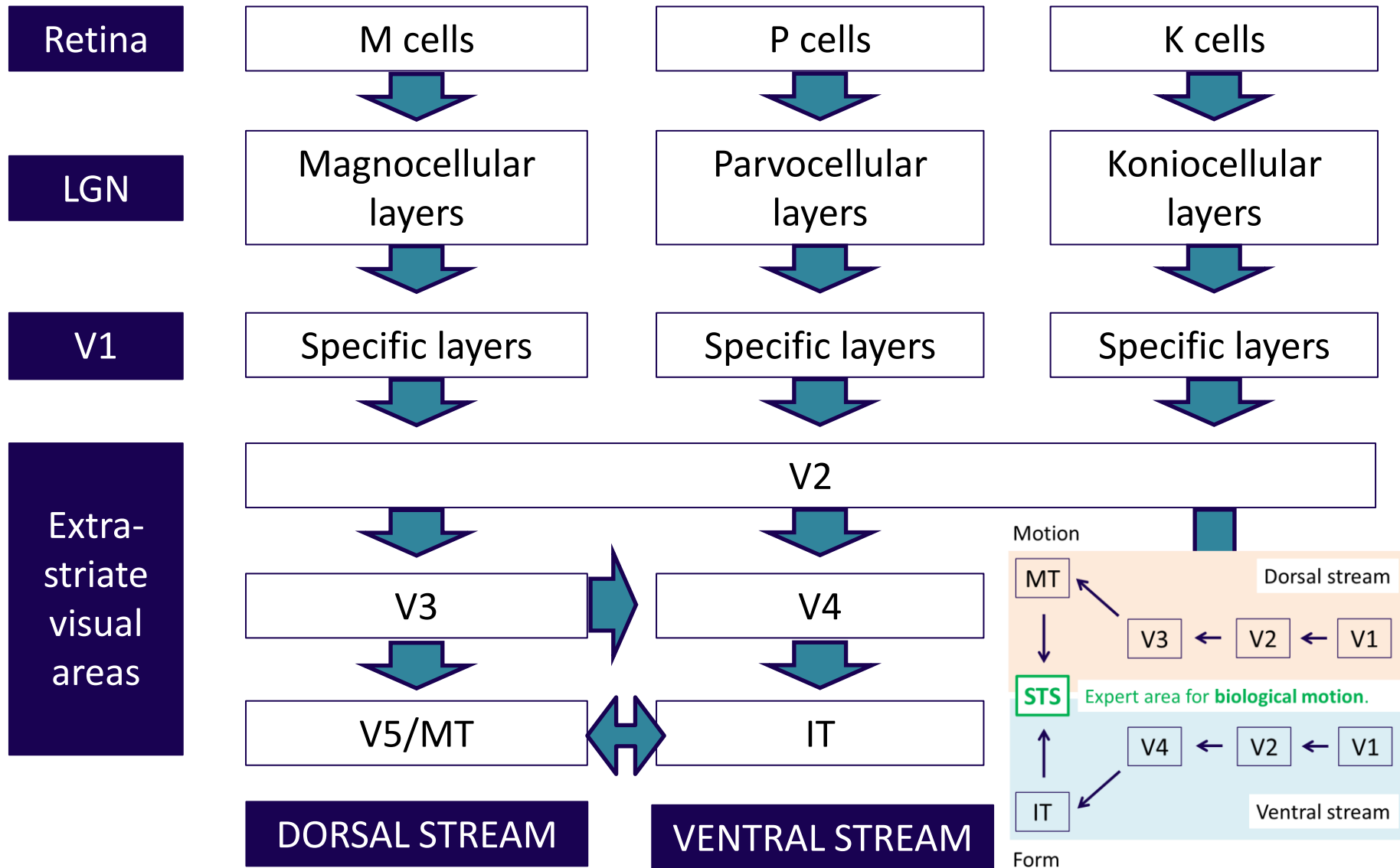
Cortical visual pathways



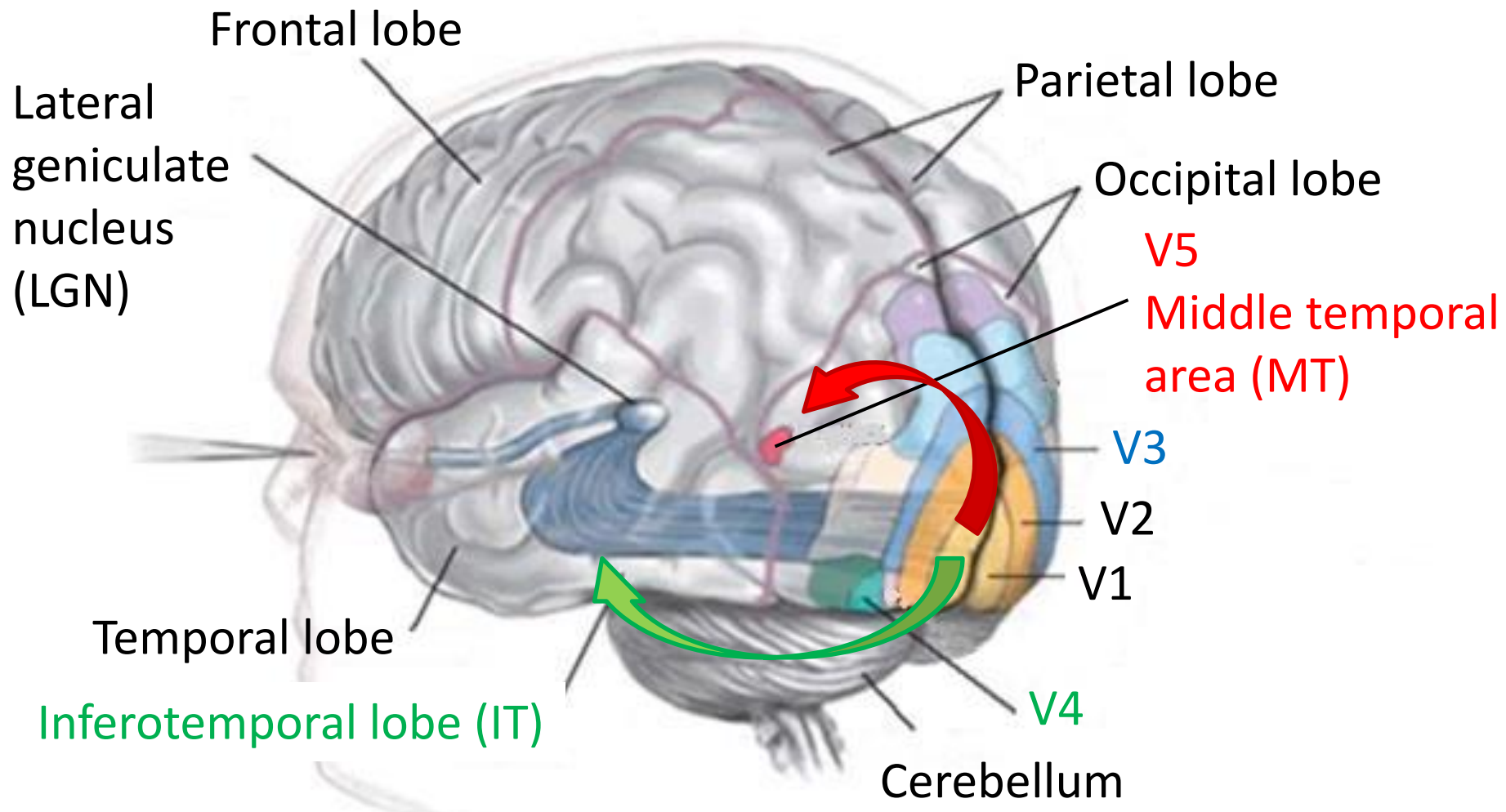
Cortical visual pathways



Cortical visual pathways



Cortical visual pathways



Temporal (ventral) stream: V1/2/4/IT WHAT stream: Object identification

Parietal (dorsal) stream: V1/2/3/MT WHERE stream: Object localisation

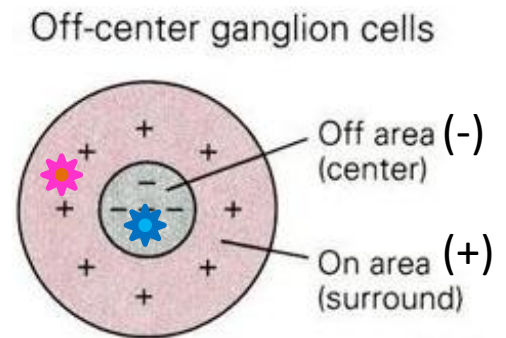
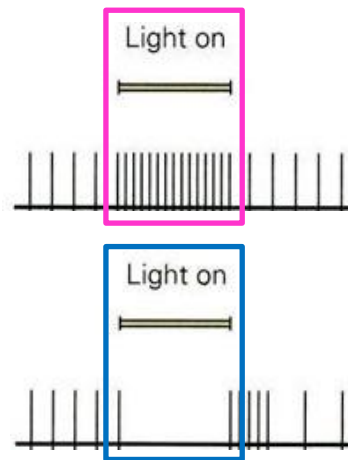
2. Specific neurons and their receptive fields

Specific neurons and their receptive fields

- **Ganglion cells** in the retina have on/off centre/surround receptive fields.
 - Luminance ON/OFF receptive fields.

Stimulus in the **excitatory area** of the receptive field **increases** neuronal firing.

Stimulus in the **inhibitory area** of the receptive field **decreases** neuronal firing.

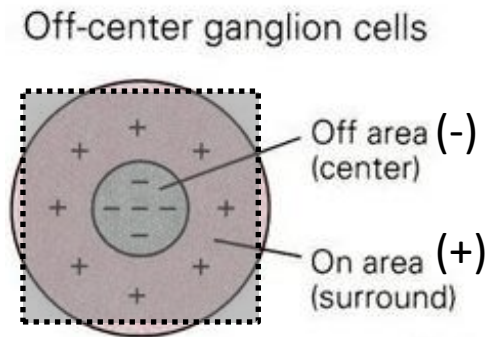
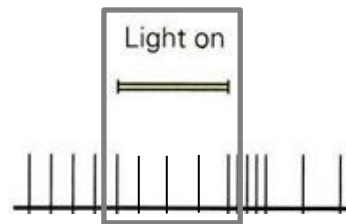


Vice versa for on-centre ganglion cells.

Specific neurons and their receptive fields

- **Ganglion cells** in the retina have on/off centre/surround receptive fields.
 - Luminance ON/OFF receptive fields.

An **object surface (uniform illumination)** covers the **excitatory AND inhibitory area** of the receptive field and does not modify neuronal firing, because (+) and (-) cancel each other out.



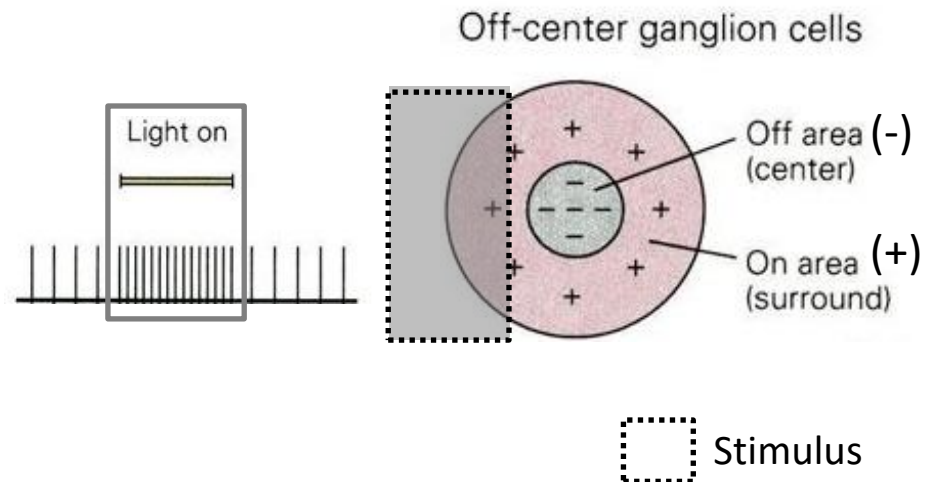
 Stimulus

Specific neurons and their receptive fields

- **Ganglion cells** in the retina have on/off centre/surround receptive fields.
 - Luminance ON/OFF receptive fields.

An **object edge** (luminance **discontinuity**) only stimulates the **excitatory surround** of the receptive field and increases neuronal firing.

i.e., supports **object segmentation**.

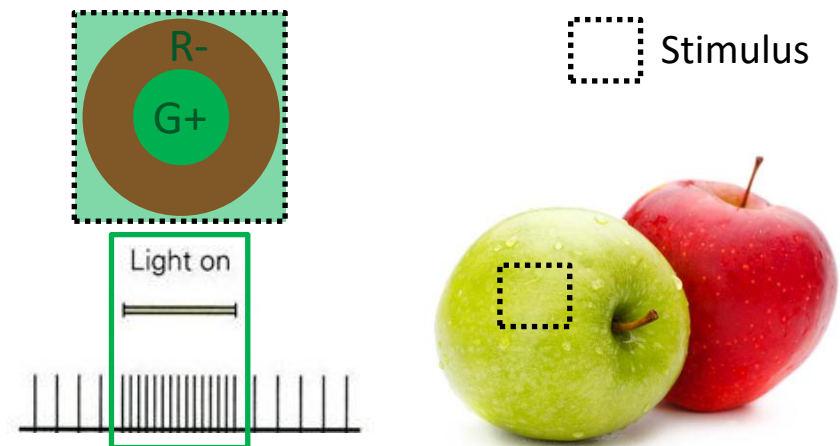


Specific neurons and their receptive fields

- **Colour-selective opponent neurons** in V1-V4 (ventral stream) have on/off centre/surround receptive fields (single-opponent receptive fields).
 - Colour ON/OFF receptive fields.
 - i.e., on for one colour / off for the opponent colour.

Single-opponent neurons,
e.g., **green-on-centre/red-off-surround**,
support perception of **colour surfaces**.

A green surface will increase neuronal firing (stimulates **green-on-centre**, leaves **red-off-surround** unaffected).

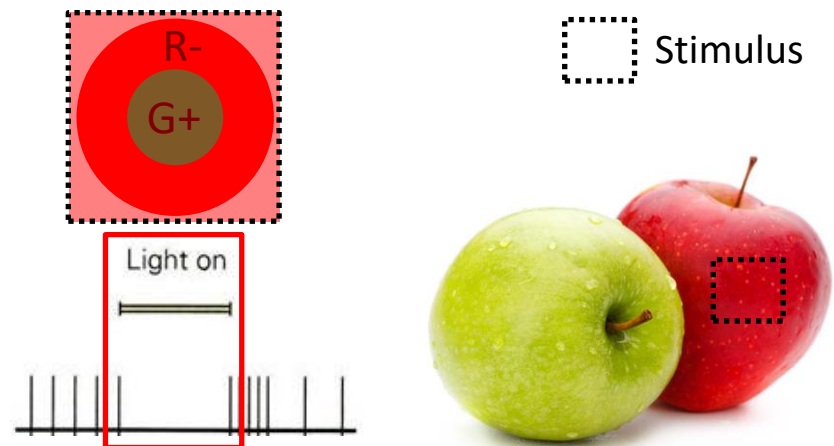


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Single-opponent neurons,
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A red surface will decrease neuronal firing (stimulates **red-off-surround**, leaves **green-on-centre** unaffected).

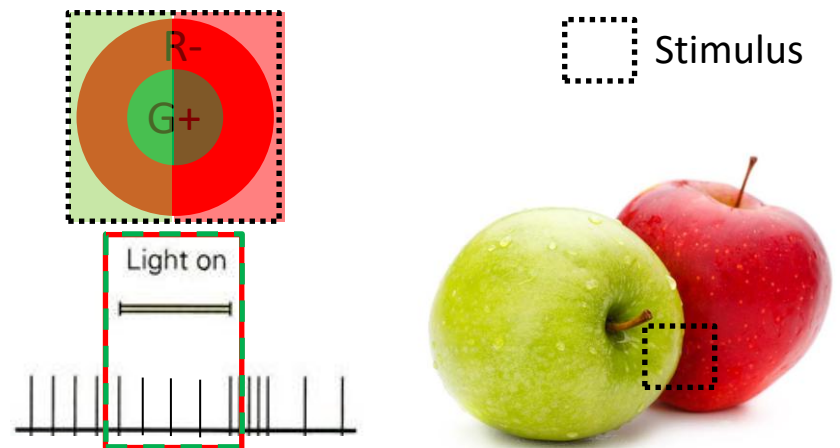


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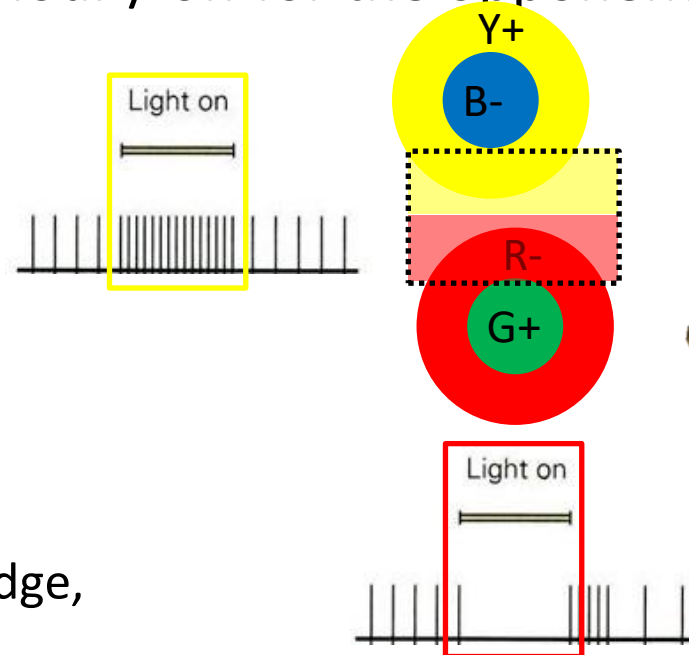
Single-opponent neurons, e.g., **green-on-centre/red-off-surround**, support perception of **colour surfaces**.

A colour edge does not affect neuronal firing (stimulates **red-off-surround** AND **green-on-centre** simultaneously).



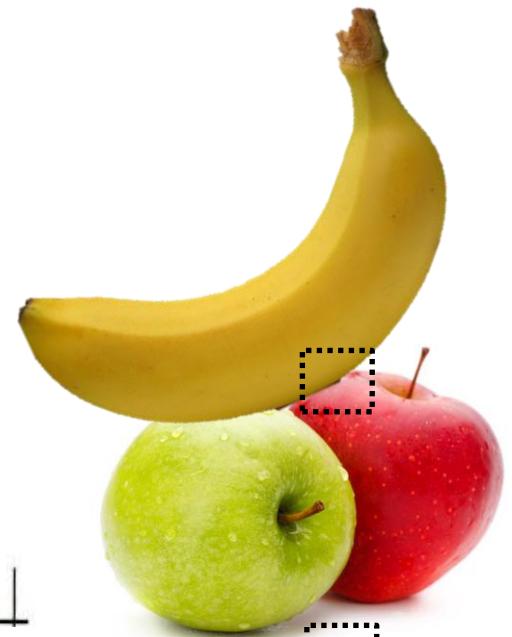
Specific neurons and their receptive fields

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Unless:

It's a non-opponent colour edge,
of course.



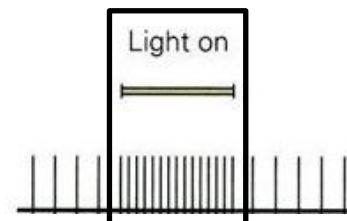
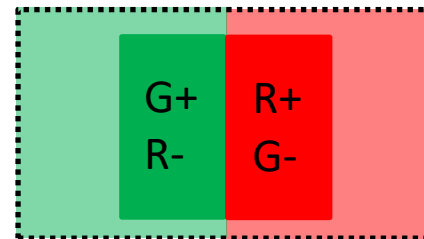
Stimulus

Specific neurons and their receptive fields

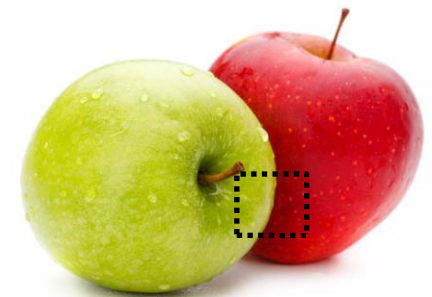
- **Colour-selective opponent neurons** in V1-V4 (ventral stream) can also have side-by-side receptive fields (double-opponent receptive fields).
 - Colour ON and OFF receptive fields.
 - i.e., on for one colour / off for the opponent colour on one side AND vice versa on the other side.

Double-opponent neurons, e.g., **green-on-red-off-side** / **green-off-red-on-side**, support perception of **colour edges (colour discontinuity)**.

An exciting colour edge increases neuronal firing (stimulates **red-on-side** AND **green-on-side**).



 Stimulus

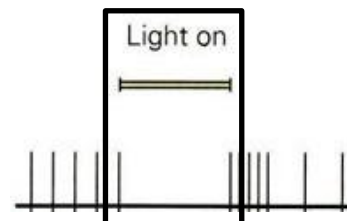
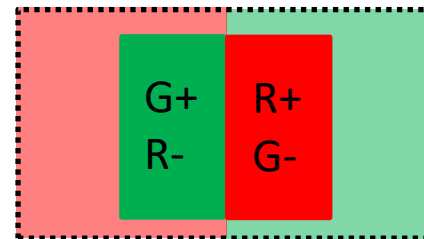


Specific neurons and their receptive fields

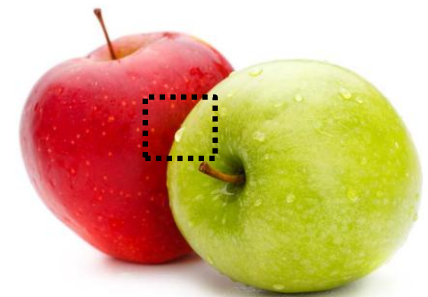
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Double-opponent neurons, e.g., **green-on-red-off-side** / **green-off-red-on-side**, support perception of **colour edges (colour discontinuity)**.

An inhibitory colour edge decreases neuronal firing (stimulates **red-off-side** AND **green-off-side**).



 Stimulus

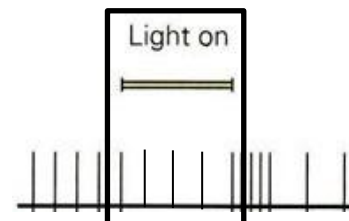
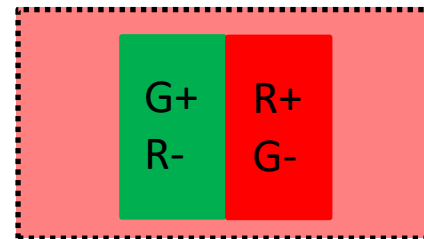


Specific neurons and their receptive fields

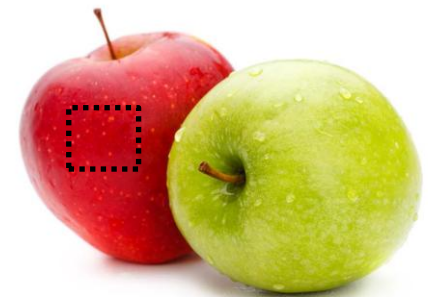
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Double-opponent neurons, e.g., **green-on-red-off-side** / **green-off-red-on-side**, support perception of **colour edges (colour discontinuity)**.

A colour surface does not affect neuronal firing (stimulates **red-on-side** AND **red-off-side**).

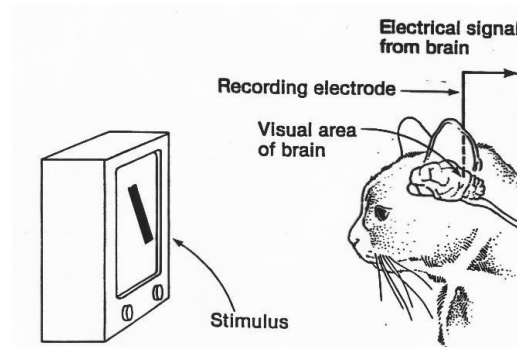


 Stimulus



Specific neurons and their receptive fields

- **Orientation selective neurons** in V1-V4 (ventral stream) have side-by-side receptive fields.
 - Neuronal response to stimuli that match the preferred orientation.



Single-cell recording from a **V1 neuron** of a cat.

Hubel & Wiesel, 1959

 A neuron's preferred orientation.

The less the stimulus matches the preferred orientation, the lower is the neuron's firing rate.



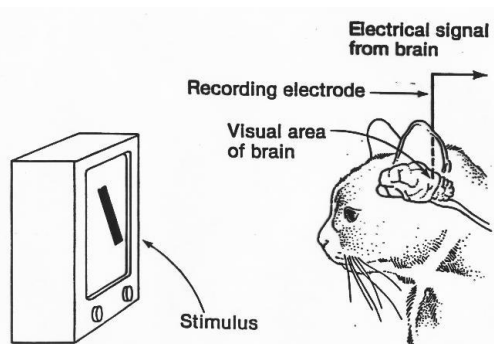
Stimulus

Neuron's firing rate

Specific neurons and their receptive fields

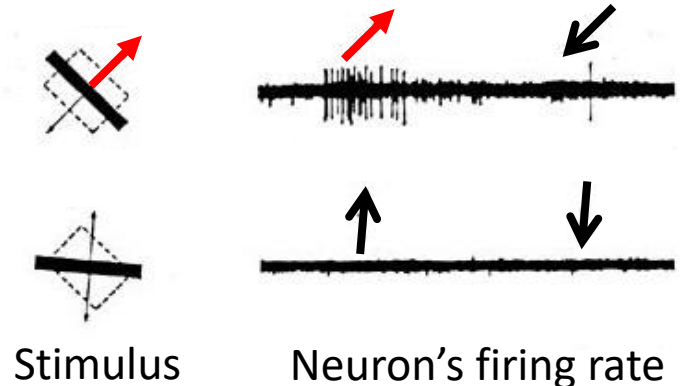
- **Motion direction selective neurons** in V1-V5 (dorsal stream) have side-by-side receptive fields.
 - Neuronal response to stimuli that match the preferred motion direction.

A neuron's preferred motion direction.



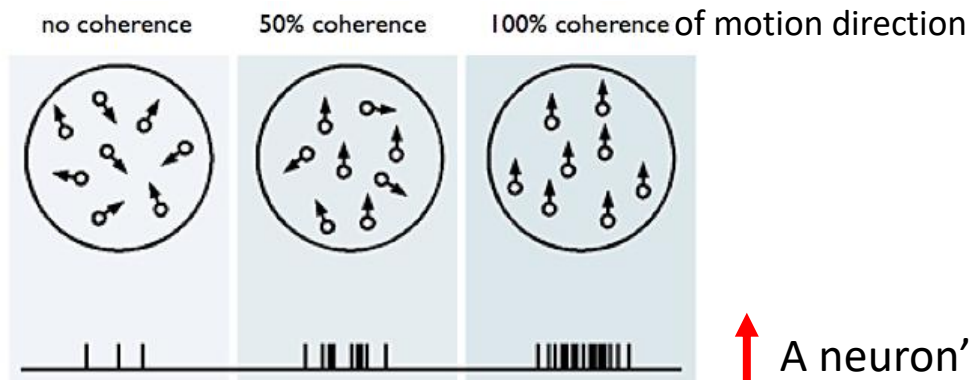
Single-cell recording from a **V1 neuron** of a cat.

Hubel & Wiesel, 1959



Specific neurons and their receptive fields

- **Motion direction selective neurons** in V1-V5 (dorsal stream) have side-by-side receptive fields.
 - Neuronal response to stimuli that match the preferred motion direction.



Single-cell recording
from a **V5 neuron** of a
monkey.

Newsome et al., 1995

↑ A neuron's preferred motion direction.

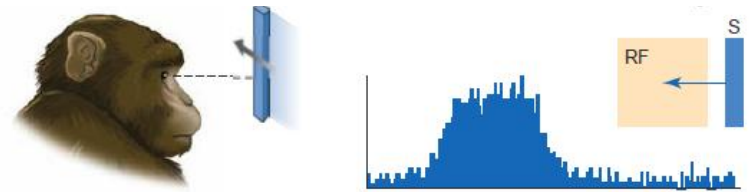
Specific neurons and their receptive fields

- **Real motion direction selective neurons** in V3-V5 (dorsal stream) have side-by-side receptive fields.
 - On for real motion, off for illusory motion.

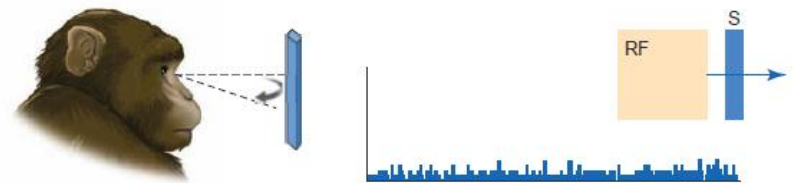
Single-cell recording from a **V3 real-motion neuron** of a monkey.
Galletti & Fattori, 2003

Real motion neurons only fire when real motion has occurred in their receptive field (they don't if retinal motion is caused by eye movements).

Real motion neurons receive a **corollary discharge signal**.



Real motion: Stimulus moves across RF.



No real motion: RF moves over the stimulus.

3. The corollary discharge theory

Corollary Discharge Theory

- Basic idea

- Our brain knows that we have perceived a moving stimulus, because **real motion induces retinal motion**.

Successive receptor stimulation on the retina,
i.e., the object representation wanders across the retina.

- Problem

- **Eye movements also induce retinal motion.**

So, how does our brain know whether real motion has occurred?

- **Answer**

- Our brain takes retinal motion **as well as** eye movements into account to determine whether real motion has occurred.

Corollary Discharge Theory

Image displacement signal (IDS)

Retinal motion.

Motor signal (MS)

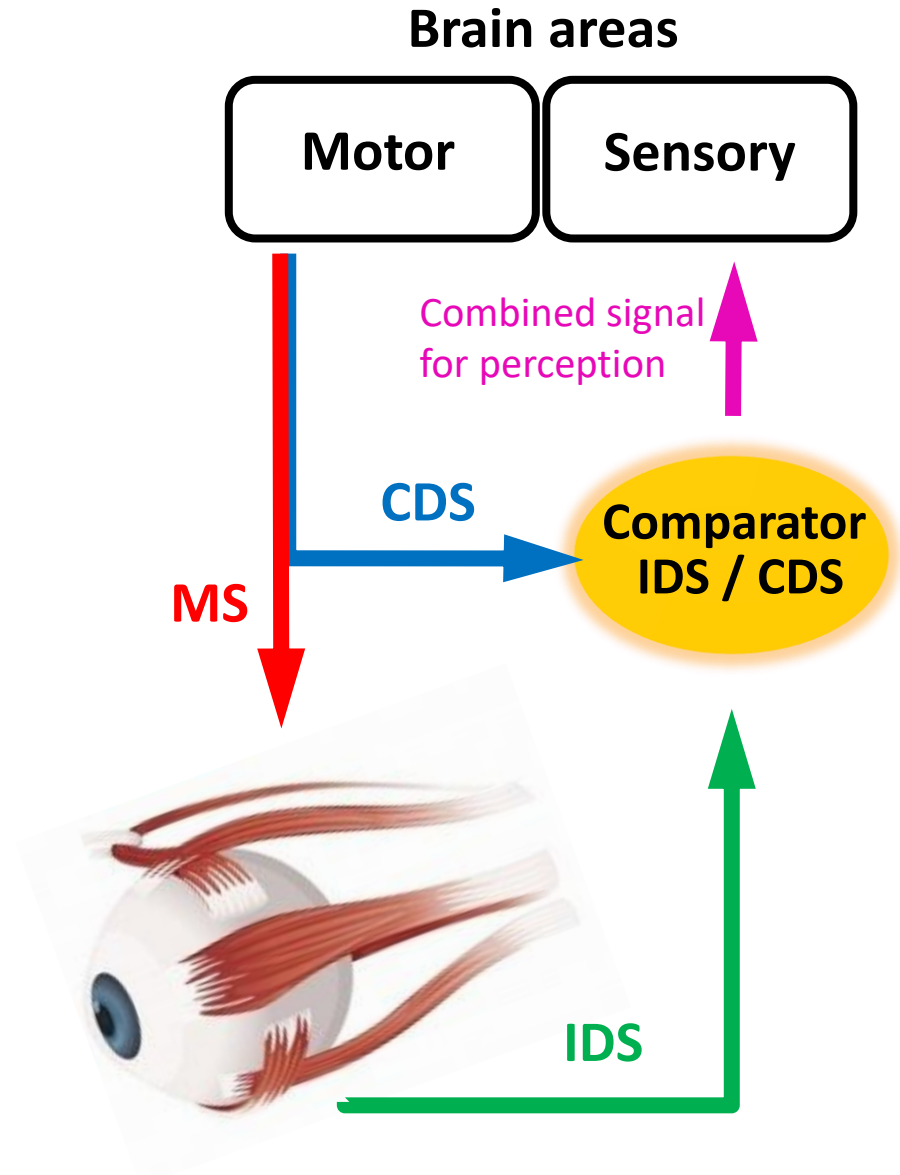
Preparation for an eye/head movement.

Corollary discharge signal (CDS)

Copy of the motor signal.

If the **combined signal** of the comparator is a **(0)**, the brain knows that **no real motion** has occurred, if it is a **(1)**, it knows that **real motion** has been observed.

There are 3 possible scenarios to test this hypothesis.



Corollary Discharge Theory

Image displacement signal (IDS) Yes (1)

Motor signal (MS) Yes (1)

Corollary discharge signal (CDS) Yes (1)

1. If the **eyes move**, but the **stimulus is static**, the comparator should forward a (0), because **no real motion** has occurred.

A



B

TRUE



Brain areas

Motor

Sensory

No real motion

(0)

(1)
CDS

Comparator
IDS / CDS
(1)/(1)

(1)

IDS

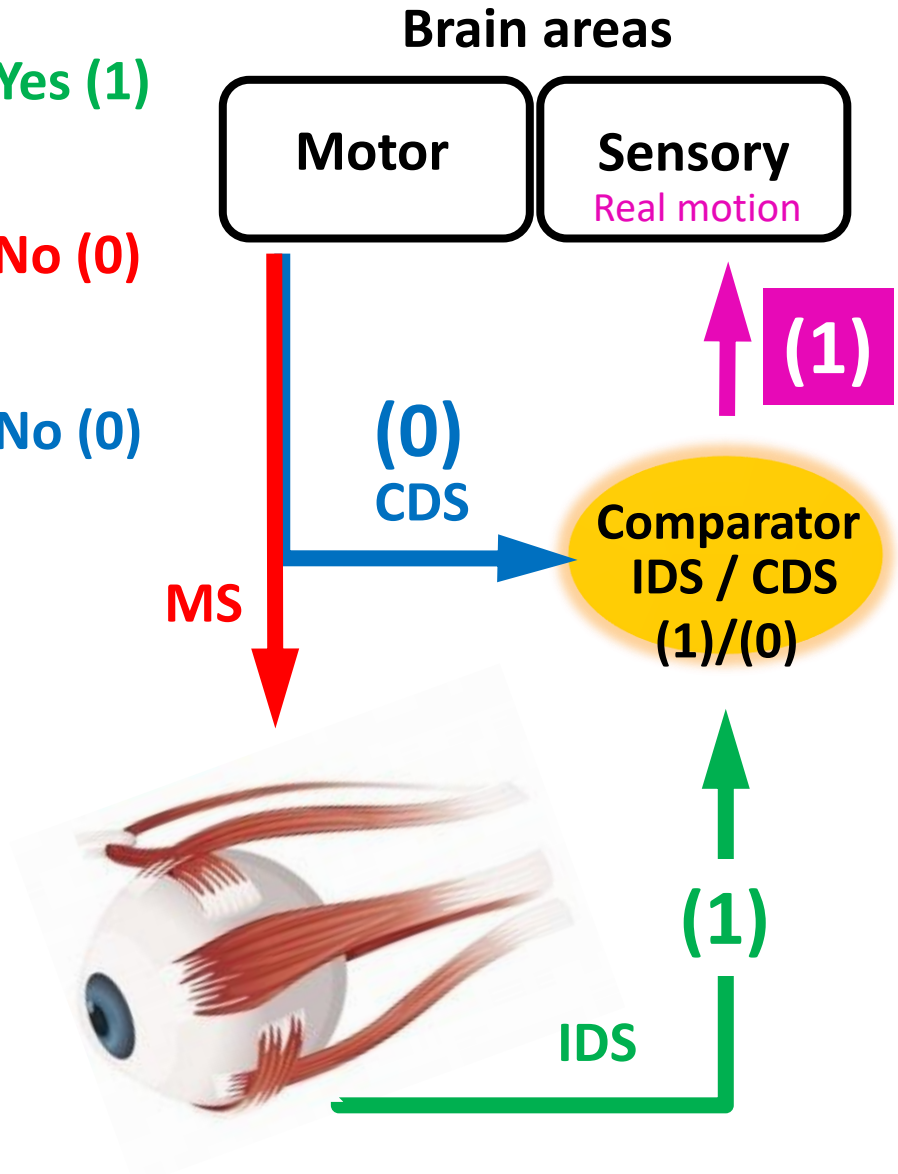
Corollary Discharge Theory

Image displacement signal (IDS) Yes (1)

Motor signal (MS) No (0)

Corollary discharge signal (CDS) No (0)

2. If the **eyes do not move**, but the **stimulus does**, the comparator should forward a (1), because **real motion** has occurred.



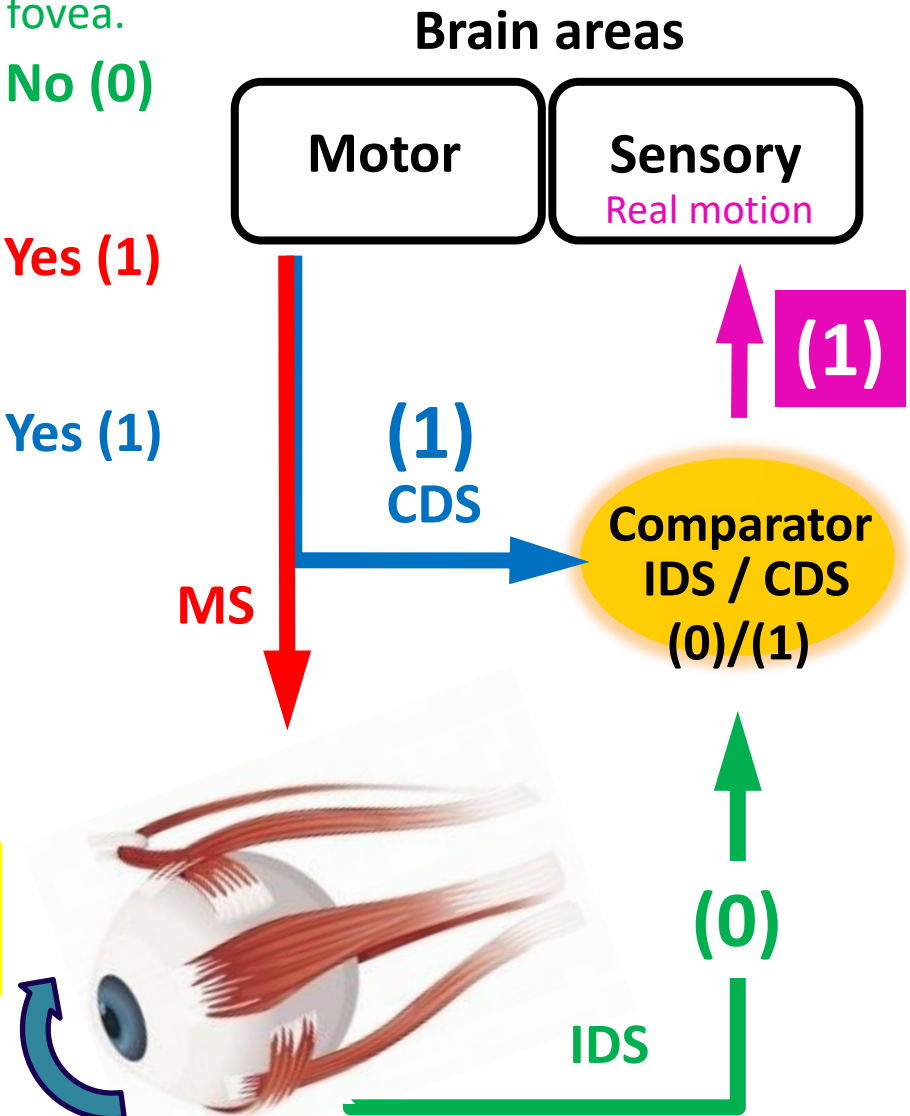
Corollary Discharge Theory

Because fixated objects are represented in the fovea.
Image displacement signal (IDS) No (0)

Motor signal (MS) Yes (1)

Corollary discharge signal (CDS) Yes (1)

3. If the **eyes move** and the **stimulus too**,
the comparator should forward a (1),
because **real motion** has occurred.



Corollary Discharge Theory

Question: What if the eyes move but the stimulus moves faster?

Don't take the 0/1 scaling too literally (used for illustration purposes). In reality, perception is rarely a categorical absent/present process but an accumulative signal that reaches a perception threshold, i.e., when the stimulus moves faster than the eyes a large part of the stimulus will be represented in the fovea and produce an IDS too small to reach the threshold.



Corollary Discharge Theory

- Evaluation

Great success!

- The corollary discharge theory successfully explains why a **moving stimulus** in a static or moving eye generates motion perception, Real motion, the comparator signals a (1).

while a **static stimulus** in a moving eye does not, despite the fact that there is retinal motion (but retinal motion is induced by eye movements only).

No real motion,
the comparator signals a (0).

Perception of real (self) motion

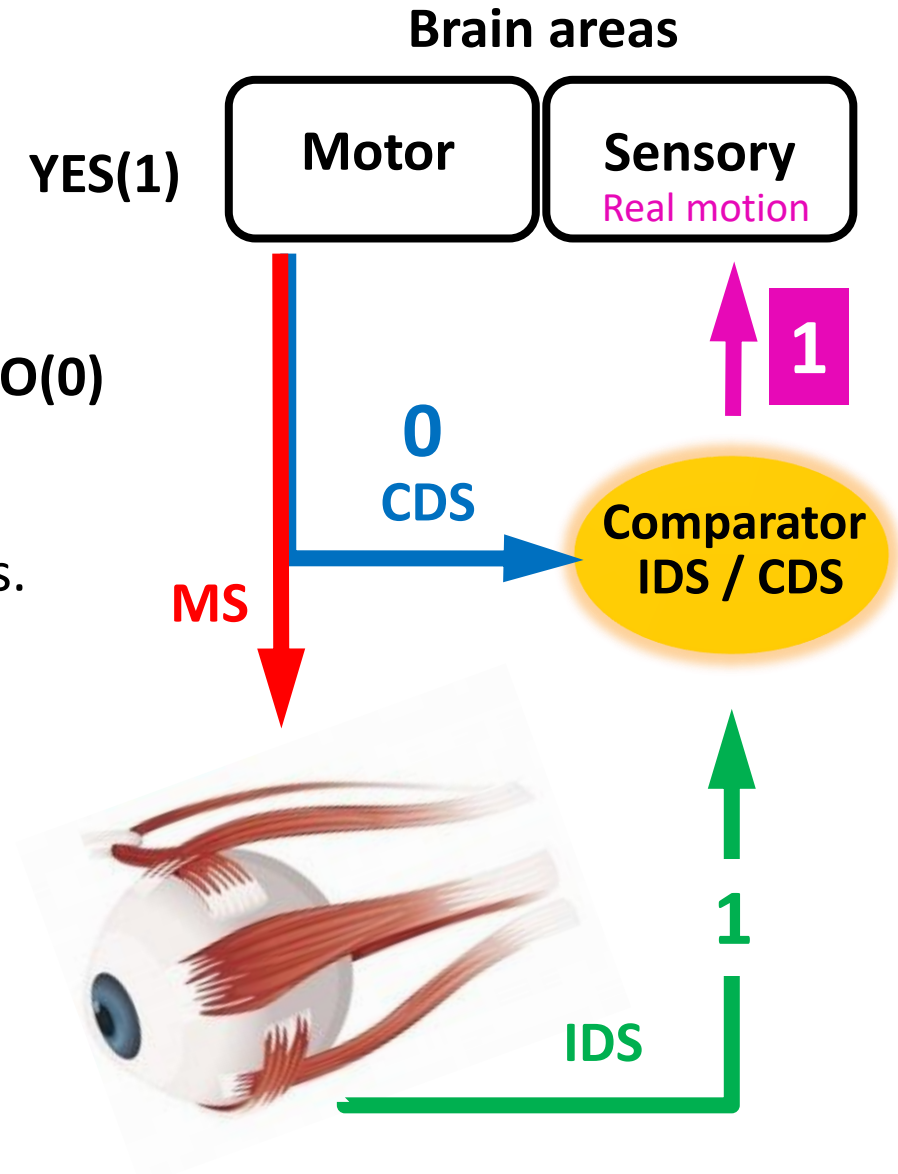
Is there...

...an image displacement signal? YES(1)

...a motor signal? NO(0)

...a corollary discharge signal? NO(0)

Moving person with no eye movements.



Illusion of “stationary objects in motion”

Is there...

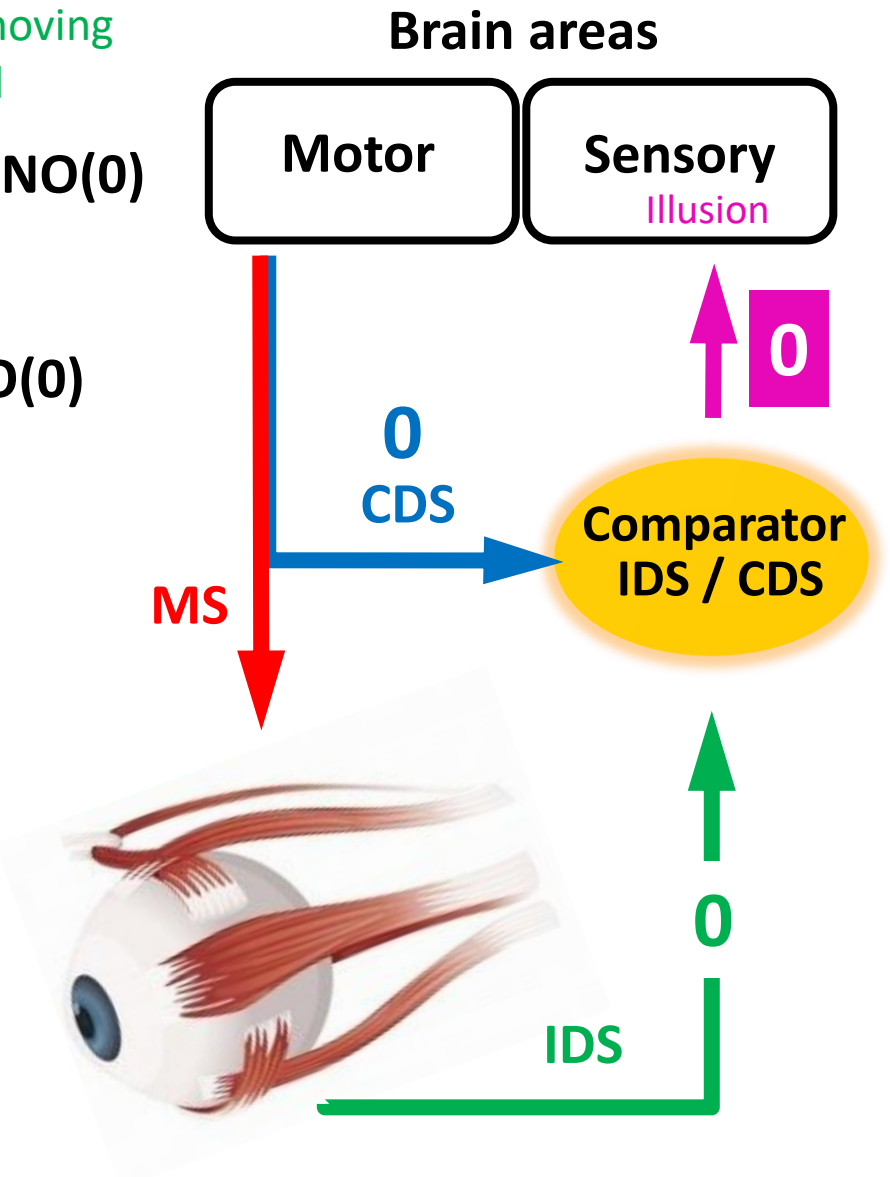
Rare – one often catches moving elements, e.g., background

...an image displacement signal? NO(0)

...a motor signal? NO(0)

...a corollary discharge signal? NO(0)

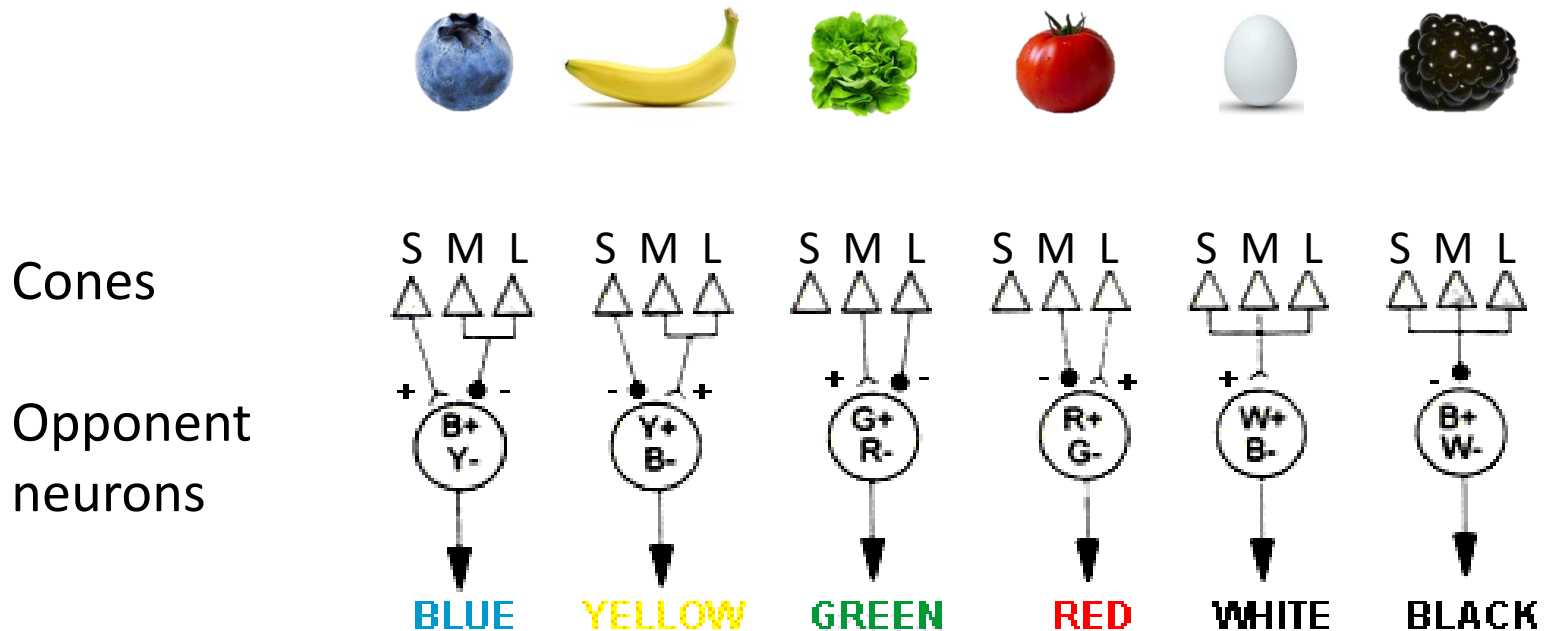
Moving person with no eye movements watching a moving object.



4. Perception of mixed colours

Perception of mixed colours

- Perception of primary colours



Perception of mixed colours

- Perception of mixed colours

Which wavelengths will be reflected from the DU logo?

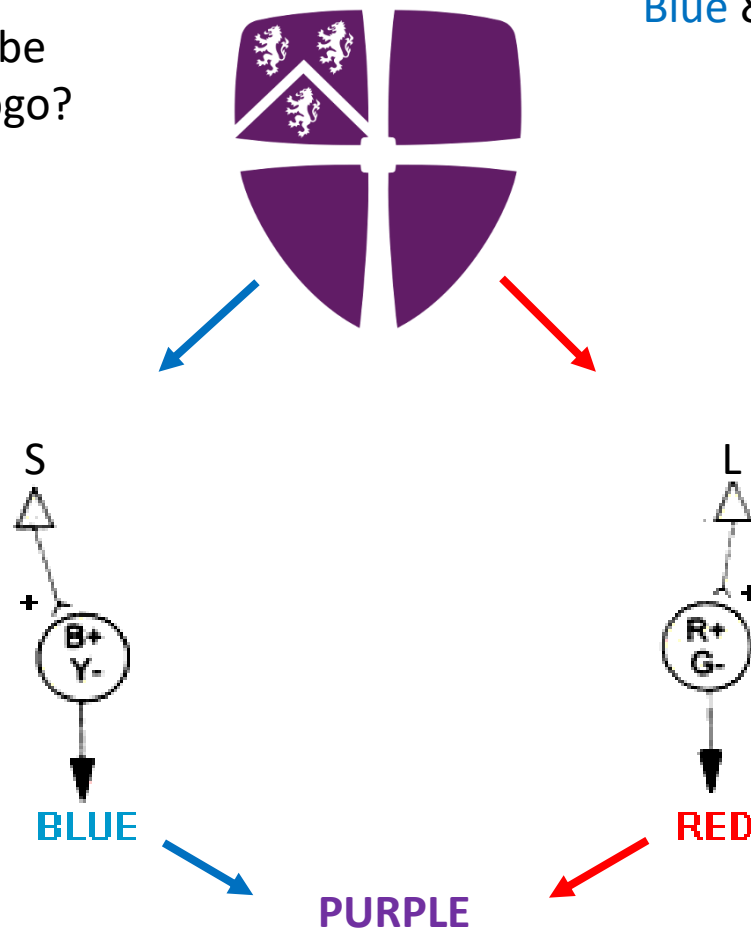
Short & Long

Which colours do we have to mix to get purple?

Blue & Red

Cones

Opponent neurons



5. Theories of perception

Bottom-up vs. top-down accounts

	Bottom-up	Top-down
Synonym	= Stimulus-guided	= Knowledge-guided



Stroop test:

<https://www.psychtoolkit.org/lessons/stroop.html>

<https://www.psychtoolkit.org/experiment-library/stroop.html>

Red
Green
Blue



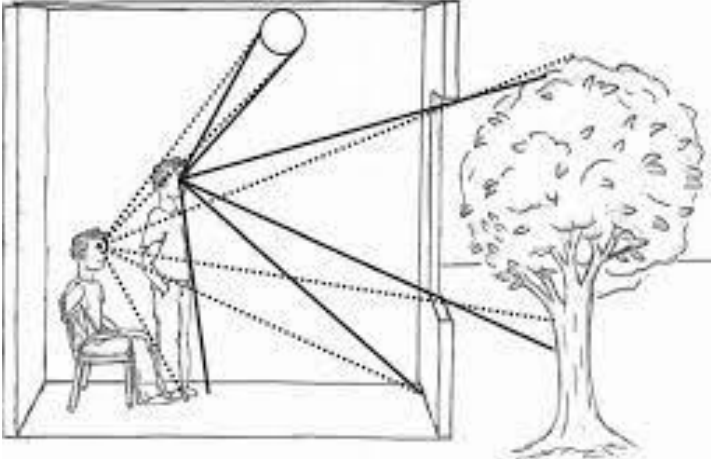
Bottom-up vs. top-down accounts

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Accounts	• Structuralism (~1900) Describing elements of consciousness by means of introspection <i>Wundt, Titchener</i> 	

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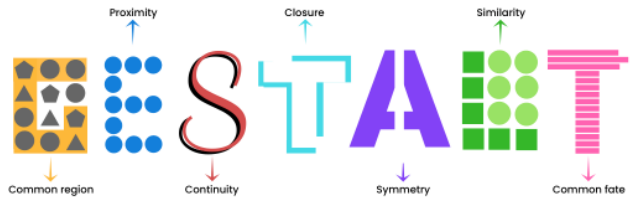
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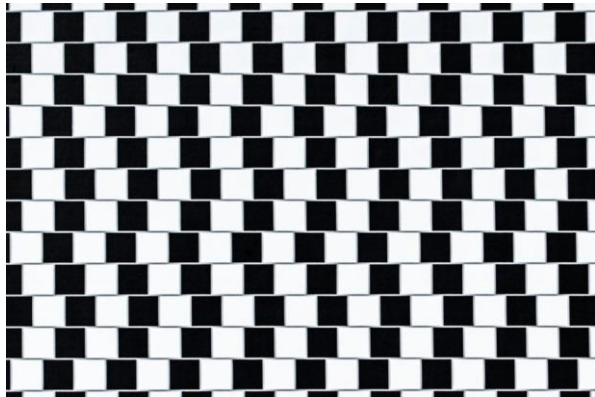
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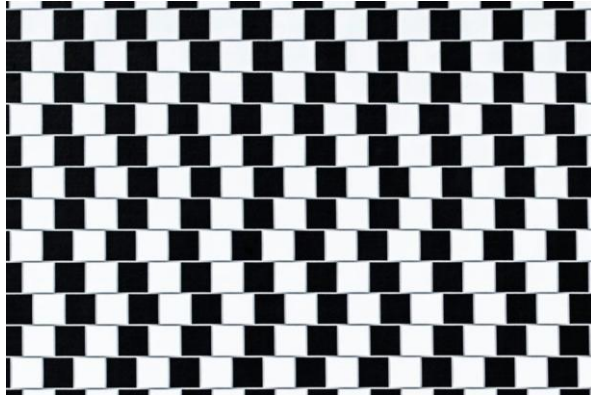
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	Bottom-up	Top-down
Basic assumptions	• Perception equals sensation and is solely determined by the retinal image, i.e., we perceive objects exactly as they are represented on the retina 	

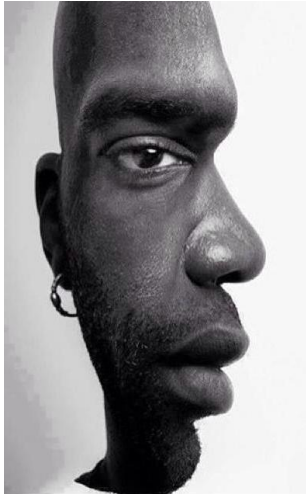
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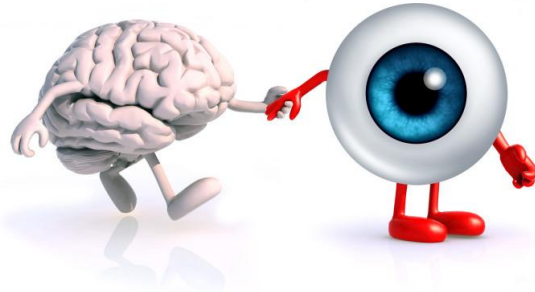
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	Perception occurs automatically (real world affordances) 	

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	• Perception occurs automatically (real world affordances)	• Perception is a question of interpretation (hypothesis testing, perceptual organization by means of Gestalt principles)
	• No cognition involved	• Cognitive processing involved
	• Perception is direct (happens in the real world) and measurable (psychophysical reaction to a stimulus)	• Perception is indirect and constructed (interpreted)

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	Perception is direct (happens in the real world) and measurable (psychophysical reaction to a stimulus)	Perception is indirect and constructed (interpreted)
	Perception is effortless, fast and always accurate (affordances don't lie)	Perception is not effortless, can be erroneous or equivocal, e.g., illusions/ambiguous figures

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	No cognition involved	Cognitive processing involved
	Perception is direct (happens in the real world) and measurable (psychophysical reaction to a stimulus)	Perception is indirect and constructed (interpreted)
Main critique	Perception is effortless, fast and always accurate (affordances don't lie) Just not true!	Perception is not effortless, can be erroneous or equivocal, e.g., illusions Why do we not learn?

More MCQs for you to practise...

- **Formative** class test
 - The test will be available from now (Tue Dec 9th) until Friday (Dec 12th, 4pm).
 - On Friday, I will reveal and discuss the correct answers.
 - All questions and answers will be made available to you on Blackboard after the lecture on Friday.
- **Revision** lecture in May
 - Quiz-style lecture with many more MCQs

See you on Friday!!

If you need a little distraction from heavy revising,
look at the wonderful visual illusions in Jain's video clip!



<https://www.youtube.com/watch?v=KDXOzr0GoA4>